

M&A as a Transformer of the Automotive Industry

This case collection explores how the automobile industry has been fundamentally shaped by mergers and acquisitions, broadly defined to include not only traditional takeovers and mergers, but also strategic alliances, joint ventures, equity participations, spin-offs, and IPOs—i.e., all transactions that materially alter corporate governance and control. Across different periods and regions, such transactions have served as key instruments for technological transformation, market expansion, restructuring, and, in some cases, value destruction. The objective of the seminar is to analyze these transactions from a financial economics perspective, focusing on the economic logic of each case and deriving generalizable insights.

Students will be randomly assigned to clusters, with the number of students in each cluster corresponding to the number of cases. Within each cluster, students are expected to self-organize as a team and coordinate both the presentation and the written output.

Each team will produce one joint term paper of approximately 6 pages per student. Individual contributions must be clearly indicated within the paper. Although each case is assigned to an individual student, the objective is not to produce separate case descriptions, but a comparative analysis across cases. Students within each cluster must explicitly compare the cases along key dimensions (e.g. governance structure, capital allocation, risk distribution, value creation) and derive two to three clear, generalizable insights from this comparison. All presentations should be framed as partial answers to a common underlying question (e.g. “Which ownership structure is best suited to manage technological uncertainty?”). Avoid pure storytelling. Descriptive summaries are necessary but not sufficient. No comparison means no insight. Even though cases are assigned individually, teams are collectively responsible for producing a coherent and integrated argument.

The presentation must not consist of a sequence of isolated case summaries. While each student will present their assigned case, the team is collectively responsible for delivering an integrated analysis. The focus must be on systematic comparison across cases and the clear presentation of overarching insights.

Students are fully responsible for their own literature research. They are expected to use advanced AI tools throughout all stages of their work, including research, structuring, analysis, writing, and validation. As an annex to the term paper, each team must provide a detailed account of the AI tools used, the prompting and workflow strategy, and their experience with these tools.

The seminar is not only about M&A, but also about the effective use of AI in a research context. All sources must be cited properly, including academic literature, data providers, and internet sources. AI-generated content must be carefully verified; hallucinations or unsupported claims are unacceptable. Hard facts, data, and well-documented evidence are essential. Quantitative support is generally more convincing than purely verbal argument. The objective is substance, not rhetoric. Content is valued, not buzzwords.

Cluster: Horizontal Mergers & Takeovers

This cluster contrasts two transactions involving Chrysler that led to fundamentally different outcomes. The objective is not to describe the deals, but to explain why one destroyed value while the other created it.

Case: Daimler Chrysler merger Daimler's merger with Chrysler was presented as a "merger of equals," but is widely regarded as a value-destructive transaction. The deal combined two companies with weak strategic fit and resulted in major integration problems and governance conflicts. Key questions: Was this truly a merger—or a disguised acquisition at an inflated price? Did Daimler overpay due to managerial overconfidence ("hubris") or strategic misjudgment? To what extent was the transaction driven by empire-building rather than shareholder value? Could the deal ever have worked, or was failure inevitable given the structure?

Case: Fiat's acquisition of Chrysler Fiat's acquisition of Chrysler under Sergio Marchionne occurred under completely different circumstances: Chrysler was in financial distress, and Fiat gained control under highly favorable terms. The transaction is often viewed as a successful turnaround and value-creating deal. Key questions: Was this an industrial merger—or primarily a financial opportunity (buying a distressed asset at near-zero cost)? How did Marchionne structure the deal to maximize shareholder value? Was success driven by superior execution—or simply by better entry conditions? Who ultimately captured the value created (Fiat shareholders, creditors, the U.S. government, or others)?

Joint task: Compare the two transactions explicitly in terms of price, timing, governance, and incentives. Answer the central question: Is success in M&A primarily driven by strategic fit and execution—or by buying the right asset at the right price? Formulate 2–3 general lessons about when horizontal mergers create or destroy shareholder value.

Cluster: Diversified Conglomerates

The two cases are about the formation of diversified conglomerates in the automotive industry. What does empirical research say about conglomerates and shareholder value? Under which conditions do conglomerates emerge despite evidence of a conglomerate discount? And why did firms like Daimler and Continental pursue diversification at a time when conglomerates elsewhere were being dismantled through hostile takeovers? The cases also raise governance questions: to what extent do these developments reflect weak corporate governance that enables value-destroying expansion?

Case: Daimler-Benz under Edzard Reuter – The "Integrated Technology Group" Under Reuter, Daimler-Benz transformed itself through acquisitions and expansion into a highly diversified technology conglomerate, including AEG (railway technology, electronics), Debis (financial and IT services), and Daimler-Benz Aerospace (DASA) with Dornier GmbH and Fokker. Many of these businesses proved unprofitable and generated few synergies, leading to significant value destruction. Subsequent management largely reversed the strategy. Overall, the conglomerate phase is widely viewed as economically unsuccessful, even though parts of the aerospace activities created long-term value.

Case: Continental – From Tire Manufacturer to Conglomerate and Back Continental AG evolved through acquisitions from a focused tire producer into a broadly diversified automotive supplier. However, rising complexity and limited synergies prompted a strategic refocusing and

gradual portfolio simplification. Put bluntly: the firm appears to be moving back toward a more focused tire business.

Students should assess whether Continental AG's diversification beyond tires was strategically coherent and value-creating, and evaluate whether the subsequent refocusing reflects a rational strategic shift or a correction of earlier overexpansion. In particular, they should analyze the true drivers of the deconglomeration—such as financial distress, capital market pressure, or a reassessment of synergies—and consider the role of corporate governance and managerial incentives in both phases.

Cluster: Divestments, Spin-Offs & IPOs

The Ferrari IPO allegedly was a great success and created tremendous wealth for the owners. Porsche tried to mimic this success with questionable results. Explain the commonalities and differences. Explore the officially stated reasons and the “true” reasons not stated. Which obstacles (opposition) needed to be overcome?

Case: Ferrari IPO Ferrari was spun off from Fiat Chrysler and taken public in 2015. The Ferrari IPO is widely viewed as a major value-creating transaction, with Ferrari achieving a premium valuation as a luxury brand.

Case: Porsche IPO Volkswagen partially listed Porsche in 2022, aiming to replicate the Ferrari model. Despite strong initial valuation expectations, the outcome has been more mixed, raising questions about whether the transaction truly unlocked value.

Key questions (for both cases): What were the official justifications for the IPOs—and what were the likely underlying economic motives? Did the transactions reduce a conglomerate discount, or primarily reallocate value between stakeholders? Who actually benefited (parent company shareholders, new investors, controlling shareholders)? What governance structures were created, and how do they affect long-term value? What obstacles (e.g. political, labor, governance-related) had to be overcome?

Joint task: Compare the two transactions in terms of valuation, governance, and value allocation. Answer the central question: Do spin-offs and IPOs create value—or do they mainly redistribute it? Formulate 2–3 general lessons about when carve-outs and IPOs are value-enhancing

Cluster: The truck industry: why did separation happen so late?

Volvo sold its passenger car division relatively early, thereby creating a much more focused truck and commercial-vehicle group. Mercedes-Benz Group and Volkswagen Group maintained integrated structures for decades. Only recently did Mercedes-Benz fully spin off its truck division, while Volkswagen still retains control over Traton. This raises a more fundamental question: if separation creates value, why did it take so long—and why is it still incomplete?

Case: Volvo – Trucks vs. Cars Volvo separated its passenger car division early (later sold to Ford and subsequently acquired by Geely), creating a focused truck business. The case serves as a benchmark for an early and relatively clean separation.

Case: Mercedes-Benz Group – Cars vs. Daimler Truck Mercedes-Benz separated its truck division into an independent listed company. The move was framed as a value-enhancing step to improve focus and capital allocation, but raises the question why this was not done earlier.

Case: Volkswagen Group – Traton vs. passenger cars Volkswagen partially listed its truck division through the IPO of Traton, while retaining control. The group structure remains more integrated, raising questions about why the separation is incomplete.

Key questions: Why did Volvo separate early, while others delayed? What prevented earlier separation at Mercedes-Benz and Volkswagen? Were integrated structures inefficient, or were they rational given past conditions such as financing constraints, synergies, or governance? What changed in recent years (capital markets, regulation, technology, management incentives) that made separation attractive? Why did Mercedes-Benz pursue a clean spin-off, while Volkswagen opted for a partial IPO with retained control? Does Volkswagen's continued ownership of Traton reflect hidden synergies—or managerial and control motives?

Joint task: Compare the three cases in terms of timing, degree of separation, and underlying incentives. Answer the central question: If separation creates value, why is it often delayed—and why is it sometimes only partial? Formulate 2–3 general lessons about timing, incentives, and incomplete separation in corporate restructuring. No comparison = no insight.

Cluster: Strategic Alliances, Joint Ventures, capital-linked alliances without a full merger

Case: Nissan – Renault The alliance between Renault and Nissan under Carlos Ghosn long stood as a textbook example of an equity-based strategic alliance without full merger: deep operational integration combined with formal independence. Governance was structured through cross-shareholdings, with Renault exercising control via its equity stake. Over time, however, the asymmetric ownership structure—combined with diverging economic performance and national interests—generated increasing tensions. The case highlights the limits of capital-linked cooperation when control rights and value creation become misaligned.

Case: Porsche – Volkswagen Group From the 1970s to the 1990s, Porsche and Volkswagen cooperated in several operational projects (e.g. the Porsche 914 and later the joint development of the VW Touareg and Porsche Cayenne). This cooperation was not based on formal alliance structures or cross-shareholdings, but was stabilized through overlapping ownership and control at the level of the Porsche–Piëch family. In this sense, coordination occurred through joint ownership at the shareholder level rather than through contractual or equity-based alliance governance. From 2005 onwards, Porsche began to build a direct equity stake in Volkswagen, shifting the relationship from coordination under shared ownership to a formal control strategy, which ultimately led to a takeover attempt and its reversal.

Case: Toyota – Subaru The relationship between Toyota and Subaru represents a form of equity-based cooperation with limited formal control. Toyota holds a minority stake in Subaru and collaborates on selected platforms (e.g. BRZ/GR86), while both firms remain legally and strategically independent. Despite Toyota's economic dominance, governance has remained relatively stable and largely conflict-free. This stability cannot be explained by ownership structure alone. Rather, it reflects Toyota's long-term governance orientation and the continued influence of its founding family, which support credible commitment and reduce incentives for opportunistic behavior in a setting characterized by incomplete contracts. The case illustrates that minority-based alliances can

be stable, but only when supported by governance structures and time horizons that align incentives beyond formal control rights.

Key questions: Why do car manufacturers such as Toyota enter into (sometimes reciprocal) minority shareholdings? What are the main drivers behind a variation in the employed approach and strategy of control, such as for the case of Porsche? What risks inherent to the ownership structure between Nissan and Renault did the economic and political environment contribute to highlight or realize?

Joint task: Compare the three cases in terms of ownership structure and their shift over time. Answer the central question: What advantages and disadvantages did the partners derive from the different collaborations and what incentives does each control strategy generate? Formulate 2-3 general lessons on the main trade-offs that different forms of collaboration between corporates expose and how these are reflected in terms of ownership and control, risk sharing, and persistence over time.

Cluster: China strategies in the automotive industry

This cluster examines how successful automotive firms can be built in China under very different governance and financing models. The focus is not on describing the companies, but on comparing alternative ways of organizing ownership, control, and capability building in a highly uncertain and rapidly evolving market. Central question: How can a successful automotive company be built in China? Which governance and financing structure is superior under which conditions—and which model is most likely to create long-term shareholder value?

Case: Geely Geely followed an M&A-driven strategy, most prominently through the acquisition of Volvo Cars, complemented by further acquisitions such as Lotus and LEVC, as well as joint ventures such as Smart. The firm operates relatively independently from the state and uses acquisitions as a tool for rapid capability building and international expansion.

Case: SAIC Motor SAIC represents the joint venture model, historically embedded in state-led industrial policy. Growth was achieved through partnerships with foreign OEMs, combining market access with technology transfer under regulatory constraints.

Case: BYD BYD pursued a vertically integrated strategy, building capabilities in batteries and electric vehicles internally. The firm illustrates a model based on technological control and deep integration rather than M&A or partnerships.

Case: NIO NIO represents a capital-market- and investor-financed entry model, relying heavily on external financing and focusing on brand, software, and services. The company combines high growth ambitions with significant cash burn and limited vertical integration.

Key questions: What are the main differences between M&A-driven, joint venture-based, vertically integrated, and capital-market-driven models? How do these structures affect access to technology, capital, and scale? Which risks are borne by whom? Which model is most robust under changing market conditions?

Joint task: Compare the four cases in terms of governance, financing, capability building, and long-term viability. Answer the central question: Which model is most likely to succeed in the long run—and why? Formulate 2–3 general lessons about governance and value creation in emerging high-growth industries.

Cluster: How to build a new car company

The automotive industry is characterized by high entry barriers, including capital intensity, technological complexity, and scale requirements. Nevertheless, new entrants continue to emerge. This cluster explores how new car companies are built under very different strategic models, and the role of capital markets, partnerships, and M&A in overcoming entry barriers.

Case: Rimac Automobili Rimac developed from a small engineering-driven startup into a key player in high-performance electric vehicles and battery systems. Growth was achieved through strategic partnerships (e.g. with Porsche and Hyundai Motor Group), selective acquisitions and strategic partnerships, and access to external capital, while maintaining a significant degree of independence.

Case: Fisker Fisker pursued an asset-light strategy, relying heavily on outsourcing (e.g. contract manufacturing by Magna International) and external partners. The company aimed to minimize capital intensity, but ultimately struggled with execution and financial sustainability.

Key questions: What are the main barriers to entry in the automotive industry, and how do different firms overcome them? What roles do partnerships, outsourcing, and M&A play in building capabilities? How do different financing models (strategic investors, capital markets, asset-light approaches) affect growth and risk? Which model is most scalable and sustainable? Who ultimately captures the value created?

Joint task: Compare the cases in terms of entry strategy, capital intensity, governance, and long-term viability. Answer the central question: What is the most viable way to build a new car company in a capital-intensive industry? Formulate 2–3 general lessons about entry strategies and value creation.

Cluster: Robotics in the automotive industry – ownership, control, and geopolitics

This cluster examines how firms organize ownership and control in robotics, a key enabling technology for the automotive industry. Robotics combines high technological uncertainty, strong capital requirements, and increasing geopolitical sensitivity. As a result, firms face trade-offs between access to capital, control over critical technology, and political constraints. Central question: Which ownership structure is best suited to develop and commercialize advanced robotics—while balancing technological, financial, and geopolitical risks?

Case: KUKA – Acquisition by Midea KUKA, a leading supplier of industrial robotics to global automotive OEMs, was acquired by Midea in 2016. The transaction provided full control and access to capital, but triggered political concerns about technology transfer and foreign control of critical industrial assets. The case illustrates the benefits and risks of full acquisition in a geopolitically sensitive industry.

Case: Agile Robots – Growth through Chinese capital, cooperation, and targeted M&A

Founded in 2018 by researchers from the German Aerospace Center (DLR) in Munich, the company has scaled to more than 2,500 employees worldwide in less than a decade. The rapid growth was achieved through a combination of Chinese investors, strategic partnerships, and targeted acquisitions (e.g. Franka Emika). The firm represents a hybrid model that combines Chinese external capital while avoiding a full foreign takeover. It cooperates with Chinese partners, raising questions about strategic dependence and political acceptability. The case highlights how selective M&A and cooperation can accelerate capability building while remaining politically feasible.

Case: Boston Dynamics – Ownership Changes

Boston Dynamics has passed through multiple owners—Alphabet, SoftBank, and Hyundai Motor Group—each pursuing a different strategy (technology development, platform investment, industrial integration). Despite strong technological capabilities, commercialization has remained challenging. The case illustrates how different owners shape strategic direction and the difficulty of monetizing frontier technologies.

Key questions: How do full acquisition, hybrid structures, and changing ownership affect control, incentives, and innovation? What are the trade-offs between access to capital and retention of strategic independence? How do geopolitical constraints influence feasible ownership structures? Which type of owner is best suited to commercialize advanced robotics?

Joint task: Compare the three cases in terms of ownership structure, governance, and commercialization outcomes. Answer the central question: Which ownership model is most effective in scaling and monetizing robotics under technological and geopolitical uncertainty? Formulate 2–3 general lessons about organizing innovation in strategic industries.

Cluster : Autonomous driving – ownership models

This cluster examines how firms organize ownership and control in the development of autonomous driving technology. Given the high uncertainty, long time horizons, and massive capital requirements, different governance models have emerged, ranging from internal development to acquisitions and joint ventures.

Central question: Which ownership structure is best suited to develop and commercialize autonomous driving technology under extreme uncertainty?

Case: Ford Motor Company – Argo AI

Ford (together with Volkswagen Group) invested heavily in Argo AI, pursuing an externally funded, jointly sponsored venture structure. Despite substantial funding, the project was ultimately shut down, highlighting the risks of shared ownership without clear commercialization pathways.

Case: General Motors – Cruise General Motors pursued a more integrated approach, retaining strong control over Cruise while providing continuous internal funding. The case illustrates the benefits and challenges of internalizing high-risk technology development.

Case: Intel – Mobileye Mobileye was founded as a specialized vision-based ADAS company and became the dominant supplier of driver-assistance systems to multiple OEMs. In 2017, Intel acquired Mobileye to gain full control over a leading autonomous driving platform and to position itself in the future automotive computing stack. Unlike many competitors, Mobileye followed a step-by-step commercialization strategy, starting with revenue-generating ADAS systems and gradually moving toward higher levels of autonomy. After the acquisition, Intel combined Mobileye's software and data capabilities with its own hardware and capital resources. In 2022, Mobileye was partially re-listed via IPO, allowing Intel to retain control while accessing capital markets and external valuation. The case illustrates a hybrid model: full acquisition to secure control over a critical technology, followed by partial separation to unlock value and maintain strategic flexibility. It also highlights the importance of having a viable intermediate business model (ADAS) to finance long-term innovation in autonomous driving.

Key questions: How do joint ventures, internal integration, and acquisitions differ in terms of control, incentives, and risk allocation? Which model provides the strongest incentives for long-term investment and commercialization? Why do some approaches fail while others succeed?

Joint task: Compare the three cases in terms of ownership structure, governance, and outcomes. Answer the central question: Which ownership model is most likely to succeed in autonomous driving—and why? Formulate 2–3 general lessons about organizing innovation under extreme uncertainty.

Cluster: Software & operating systems – build, buy, or partner?

This cluster examines how automotive firms organize the development of in-car software and operating systems. Control over the software stack is increasingly seen as a key strategic asset, raising the question of whether firms should build capabilities internally, acquire them, or rely on external partners. Central question: Who should control the “operating system of the car”—and which governance model is best suited to achieve this?

Case: Volkswagen Group – CARIAD Volkswagen pursued an internal “build” strategy by establishing CARIAD as its software unit. The case highlights the challenges of developing complex software capabilities within a traditional automotive organization.

Case: Volkswagen Group – Rivian Volkswagen complemented its internal efforts through a partnership with Rivian, gaining access to external software capabilities. The case illustrates a “partner” approach to accelerate development and reduce execution risk.

Case: Stellantis – Foxconn / Amazon Stellantis relies on a hybrid model, combining multiple partnerships with technology firms. This approach reflects a more modular strategy, distributing development across external partners.

Key questions: What are the advantages and disadvantages of building software capabilities internally versus relying on external partners? How do different governance structures affect speed, control, and long-term value creation? Can traditional OEMs realistically compete with technology firms in software development?

Joint task: Compare the three cases in terms of governance, capability development, and control over the software stack. Answer the central question: Which approach—build, partner, or hybrid—is most likely to succeed in the long run? Formulate 2–3 general lessons about organizing software development in complex industrial settings.

Cluster: Battery & EV technology – investing under uncertainty

This cluster examines how firms invest in highly uncertain, capital-intensive technologies such as next-generation batteries. The focus is not on the technology itself, but on the different governance and financing structures used to manage extreme uncertainty and risk.

Central question: How should firms invest in uncertain breakthrough technologies? Which structures allow them to balance risk, control, and upside potential?

Case: Volkswagen Group – QuantumScape Volkswagen invested in QuantumScape through a minority stake, effectively creating a real option on a potentially transformative battery technology. The case illustrates how firms can gain exposure to high-risk innovation without committing to full integration.

Case: Stellantis – Battery joint ventures Stellantis entered battery production through joint ventures with partners, sharing both risk and investment. This model aims to secure access to critical technology while limiting downside exposure and capital commitment.

Case: Northvolt Northvolt represents the independent battery-platform model: a specialized company financed by OEMs, financial investors, debt, and public support, with long-term customer contracts intended to secure demand. The case is especially useful because it shows both the appeal and the risks of separating battery investment from full OEM control. Northvolt's difficulties raise questions about execution risk, overexpansion, capital intensity, customer dependence, and whether independent platforms can survive when technology, demand, and financing conditions change simultaneously.

Key questions: How do minority investments, joint ventures, and independent platforms differ in terms of risk allocation, control, and incentives? Which structure is best suited for highly uncertain technologies? How do these models affect long-term value creation?

Joint task: Compare the three cases in terms of governance, financing, and risk allocation. Answer the central question: Which investment structure best balances uncertainty, control, and value creation? Formulate 2–3 general lessons about investing in high-risk, capital-intensive technologies.